

MAT1375, Classwork6, Fall2026

Ch6. The Inverse of a Function

1. (Review) Let f and g be the functions defined by the table below. Complete the table by performing the indicated operations.

x	1	2	3	4	5	6	7
$f(x)$	<u>4</u>	5	7	0	-2	<u>6</u>	4
$g(x)$	6	-8	5	<u>2</u>	9	<u>11</u>	2
$g(x) + 3$	$6+3=9$	-5	8	5	12	14	5
$f(x) - 2g(x)$	$f(1)-2g(1)=4-2\cdot6=-8$	21	-3	-4	-20	-16	0
$g(x+3)$	$g(1+3)=g(4)=2$	9	11	2	undefined	undefined	undefined
$(f \circ g)(x)$	$f(g(1))=f(6)=6$	$f(g(2))=f(-8)$ undefined	-2	5	undefined	undefined	5
$(g \circ f)(x)$	$g(f(1))=g(4)=2$	9	2	undefined	undefined	11	2
$(g \circ g)(x)$	$g(g(1))=g(6)=11$	undefined	9	-8	undefined	undefined	-8

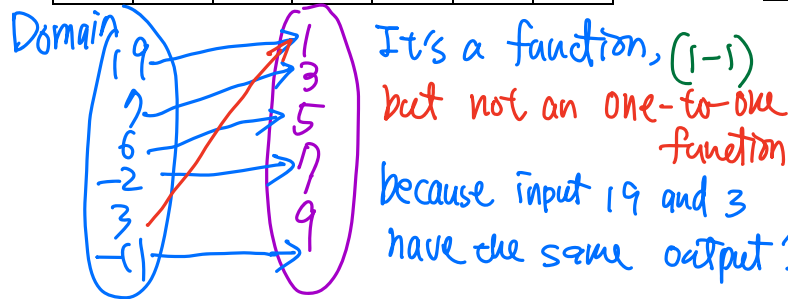
2. Complete the definition of the **one-to-one function** (or **injective**):

Given a function $f(x)$. If any two different inputs $x_1 \neq x_2$ always have different outputs $f(x_1) \neq f(x_2)$, then we call this function f a **one-to-one function**.

3. The tables below describe assignments between inputs x and outputs y . Determine which of the given tables describe a one-to one function.

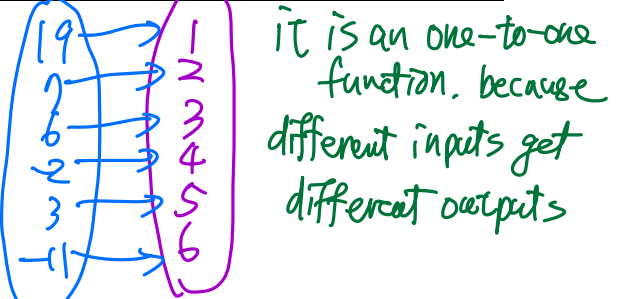
(a)

x	19	7	6	-2	3	-11
y	1	3	5	7	1	9



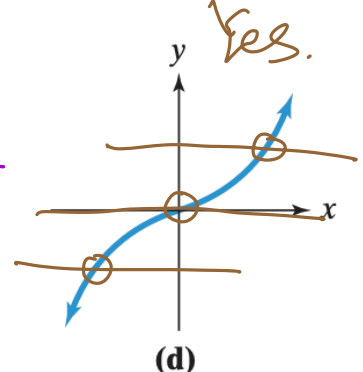
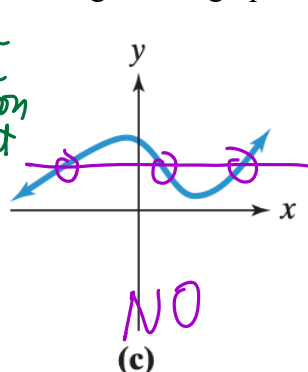
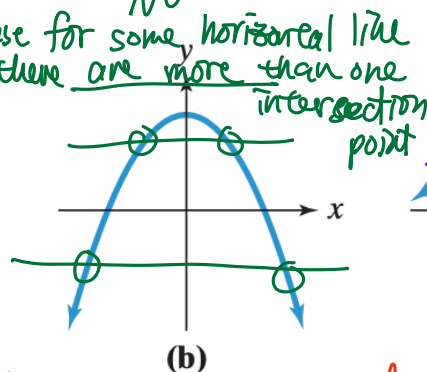
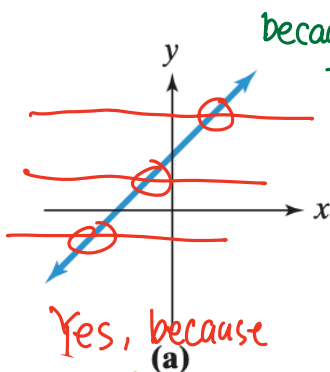
(b)

x	19	7	6	-2	3	-11
y	1	2	3	4	5	6



4. **Horizontal Line test:** A function is one-to-one when every horizontal line intersects the graph of the function at most once.

5. Use **Horizontal Line Test** to determine which of the following are the graphs of one-to-one functions.



for each horizontal line, there is only one intersection

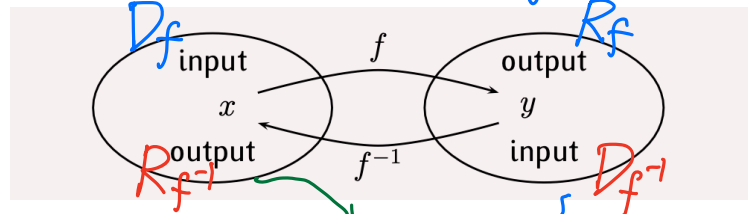
6. Complete the definition of the **Inverse of a Function**:

Let f be a function with domain D_f and the range R_f , and assume that f is one-to-one. The **inverse** of f is the function f^{-1} , determined by:

$f^{-1} \neq \frac{1}{f}$
 $\underline{\underline{f^{-1}(x)}}$

$f(x) = y$ means precisely that

$f^{-1}(y) = x$



Therefore, we have $D_{f^{-1}} = R_f$, and $R_{f^{-1}} = D_f$.

7. How to check if two given functions are **inverse** with each other:

Let f and g be two functions such that

$f(g(x)) = x$ for every x in the domain of g **and**

$g(f(x)) = x$ for every x in the domain of f .

The function g is the **inverse of the function f** and is denoted by $g = f^{-1}$.

8. How to find the inverse function for a given **invertible** function $f(x)$:

Step1: Replace $f(x)$ by y

Step2: Interchange x and y

Step3: Solve for y (isolate y)

Step4: Replace y by $f^{-1}(x)$

9. Given a function $f(x) = x^2 + 1, x \geq 0$.

(a) Find the inverse function of $f(x)$.

(b) Graph f and f^{-1} in the same coordinate system.

(a) step1 $y = x^2 + 1, (x \geq 0)$

step2 $x = y^2 + 1, (y \geq 0)$

step3 $y^2 = x - 1, (y \geq 0)$

$y = \pm \sqrt{x-1}$ $(y \geq 0)$

$y = \sqrt{x-1}$

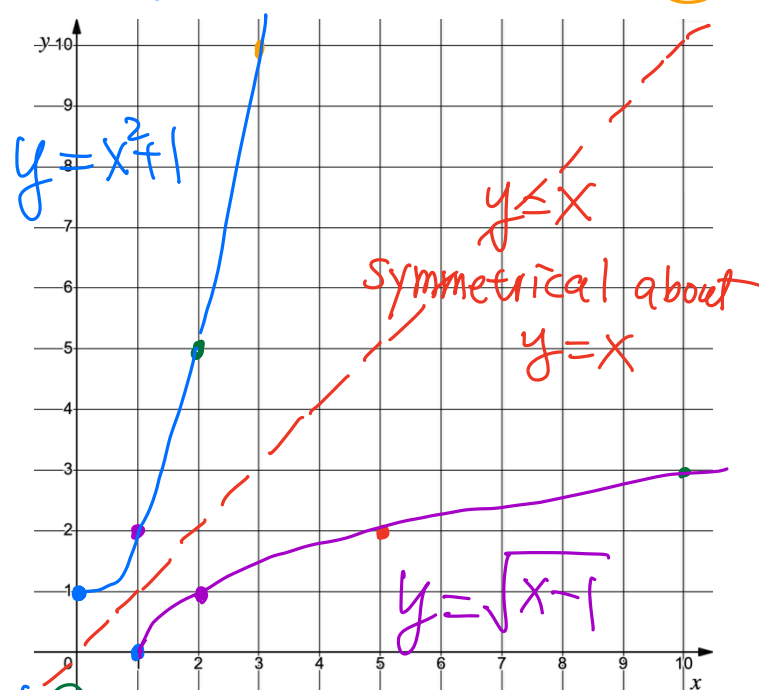
step4 $f^{-1}(x) = \sqrt{x-1}$

(b) draw $y = \sqrt{x-1}$

$y = x^2 + 1$

graphing

x	0	1	2	3
y	1	2	5	10



x	1	2	5	10
y	0	1	2	3

